ASX Announcement

25 July 2022

Positive Rare Earth Metallurgical Testwork Provides Pathway to Commercial Product

- Metallurgical testing by ANSTO using various extraction techniques demonstrates **total rare earth extraction up to 82%, averaging 76%.**
- **Extraction of high value magnet rare earths up to 86%, averaging 82%.**
- Further testwork underway to refine and improve the metallurgical processes.
- Upstream mineral processing techniques also being evaluated to upgrade mineralisation and remove non-target elements prior to extraction.

Commenting on these metallurgical results, Meeka's Managing Director Tim Davidson said: *"When evaluating clay rare earth projects, key consideration is given to scale, grade and metallurgy. Drilling has demonstrated shallow high-grade rare earths are pervasive within our 100% owned province scale land package. With metallurgical results now showing there is a pathway to production of a commercial product, we plan to advance these large scale, high-grade projects at pace.*

Drilling is planned at both Cascade and Circle Valley to progress toward delivery of a maiden Mineral Resource this financial year. Additionally, further testwork is now underway to improve on and refine the metallurgical processes. Given the tier-1 location, geopolitically critical nature of the metals and high level of permanent magnet elements present, our projects are shaping up to be significant."

Meeka Metals Limited ("**Meeka**" or "**the Company**") is pleased to report results from initial rare earth metallurgical testwork completed by ANSTO. The positive results will be used to optimise extraction options and to identify methods for material upgrade and rejection of non-target elements.

This phase of metallurgical work was completed between May and June 2022 on five composite samples from Circle Valley at various head grades (from 805ppm TREO to 1,386ppm TREO). Pleasingly, this work showed total rare earth extraction achieved up to 82%, averaging 76%. Importantly, the high value magnet rare earths leached extremely well, up to 86% extraction and an average of 82%. These results were achieved using 10% concentration hydrochloric acid and a 6-hour leach time at 50°C. Testing with sulphuric acid showed an approximate 10% reduction in recoveries compared to hydrochloric acid.

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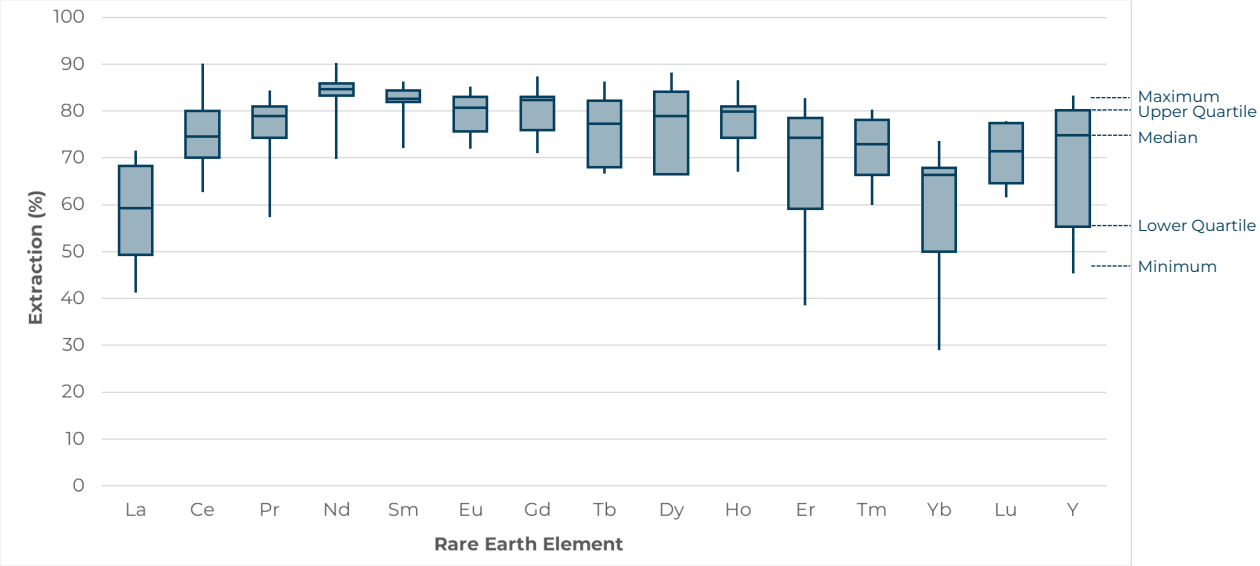


Figure 1: Extraction of rare earths using 10% concentration hydrochloric acid, 6-hour leach time at 50°C.

Further testwork is now underway to improve on and refine the metallurgical processes, optimising acid consumption and temperature with recovery. The Company is also evaluating mineral processing techniques to upgrade mineralisation and remove non-target elements prior to extraction.

Rare Earth Projects Forward Activity Plan

Current plans for advancing this significant opportunity in rare earths are shown below.

Table 1 – Rare Earth Projects Forward Activity Plan for FY23

	Sept Qtr FY23	Dec Qtr FY23	Mar Qtr FY23	Jun Qtr FY23
Circle Valley		Resource Definition Drilling	Resource Definition Drilling	**Mineral Resource Estimate**
Cascade	Central Block Drilling*	Northern Block Drilling		
Mineralogy & Metallurgy		Mineralogical Analysis	Upstream Mineral Processing Assessment	

- **Cascade Activity Plan**
  - Broad spaced (~1km between holes) drilling across the northern (525km^2) and central (400km^2) blocks.

Surface samples taken within these blocks returned grades up to 633ppm TREO from the depleted zone. The potential for significant zones of clay hosted rare earth mineralisation below these high-grade surface samples is considerable. The planned drilling will guide further programs focussed on the highest value (high heavy and magnet REO content) zones of mineralisation, ultimately targeting an initial Mineral Resource. Samples will also be taken to further refine and optimise metallurgical processes.

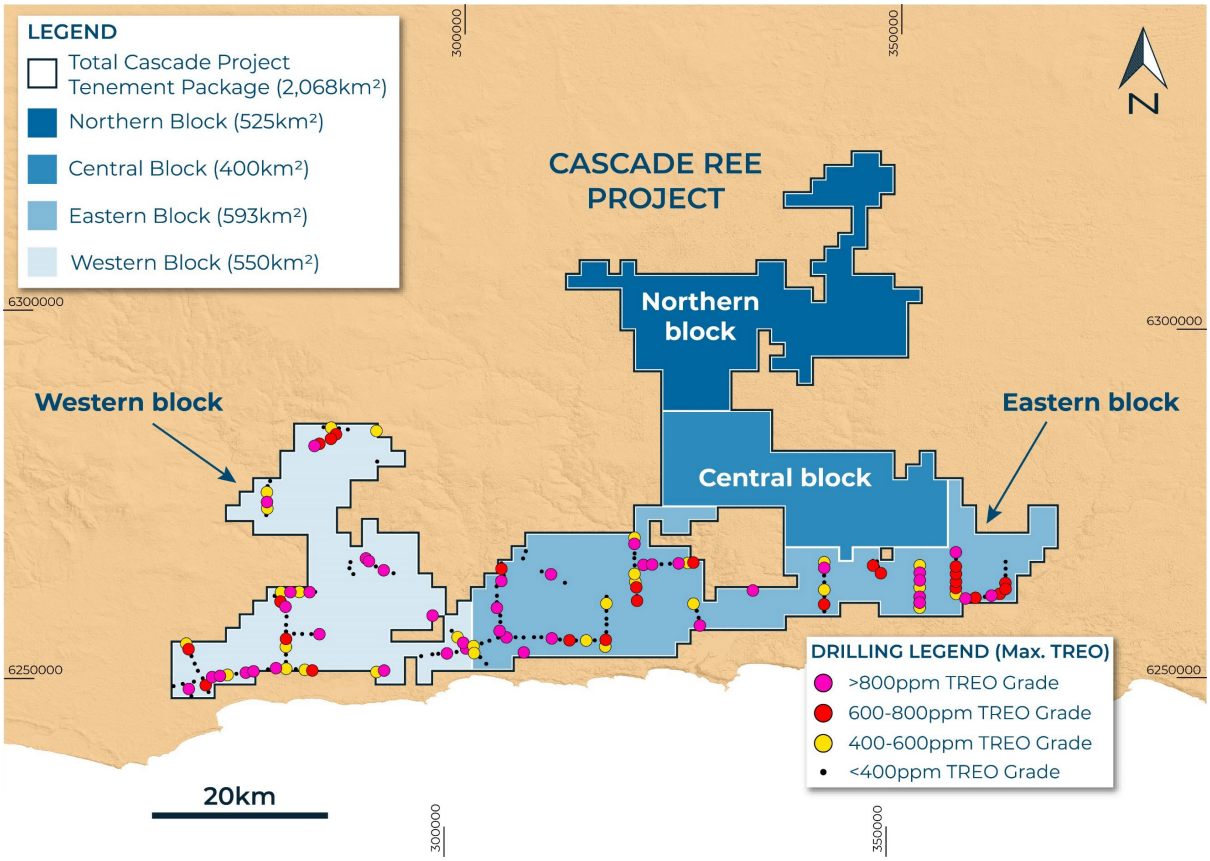


Figure 2: Meeka's 100% owned Cascade Rare Earths Project (2,068km2) showing the location of the northern and central blocks where high-grade REE is present in surface sample and drilling is planned to commence in September 2022.

- **Circle Valley Activity Plan**
  - Targeted infill drilling of high-grade zones of mineralisation.
  - Estimation of an initial Mineral Resource for Circle Valley in the first half of 2023.

  Drilling completed in the first half of 2022 identified large kilometre scale zones with thick, high-grade rare earth mineralisation. The mineralisation is located within the saprolite clay horizon, which blankets the project in varying thickness from 2m to over 50m. Assays demonstrate the pervasive nature of the high-grade rare earths, up to 36m thick and grading up to 1,748ppm TREO. The mineralisation also persistently demonstrates a high proportion of the grade, up to 47%, as valuable magnet rare earths.

  While 14,957m of samples from the 2022 drill program are pending assay (anticipated to be reported in July/August 2022) there is sufficient information now available to plan for Mineral Resource definition drilling. The planned drilling will infill zones of mineralisation and form the basis for a Mineral Resource to be completed in the first half of 2023.

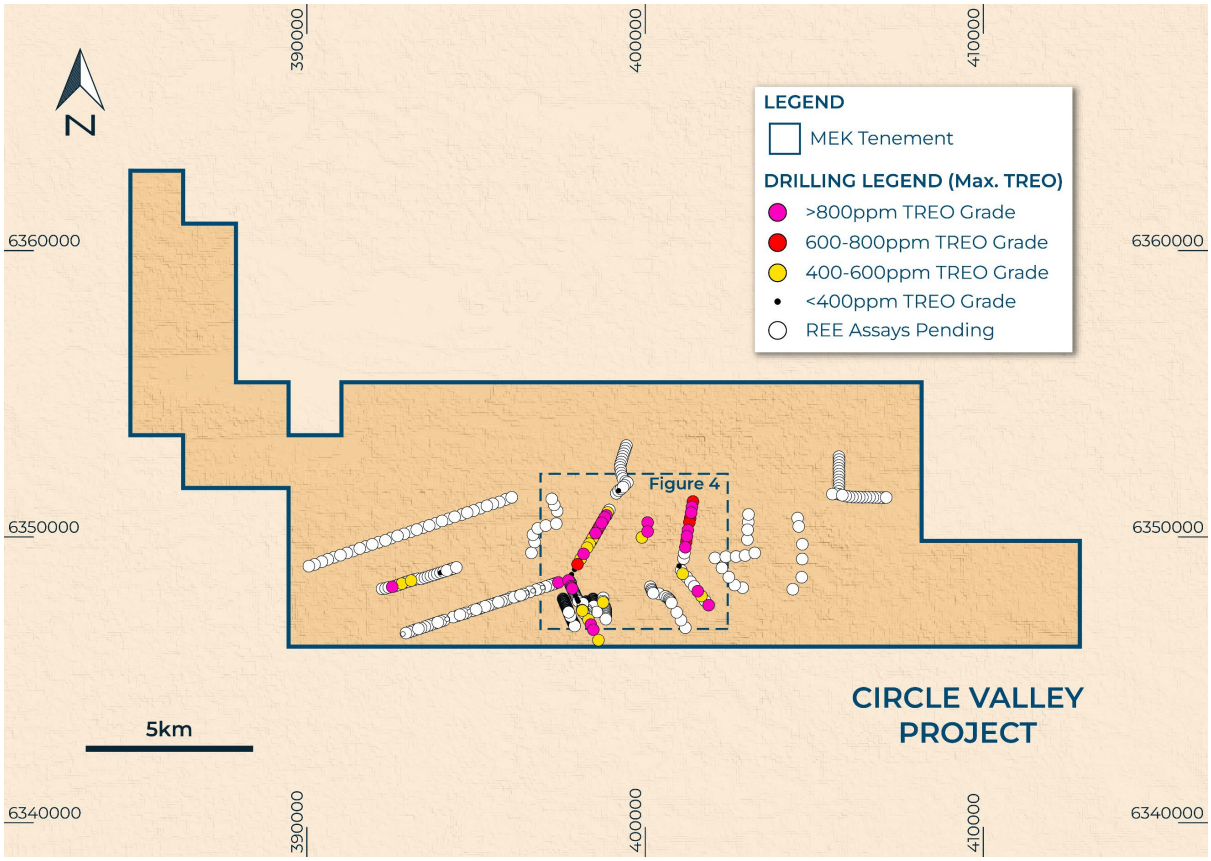


Figure 3: Meeka's 100% owned Circle Valley Project (222km2) showing collar locations, holes for which assays have been received and holes with assays pending.

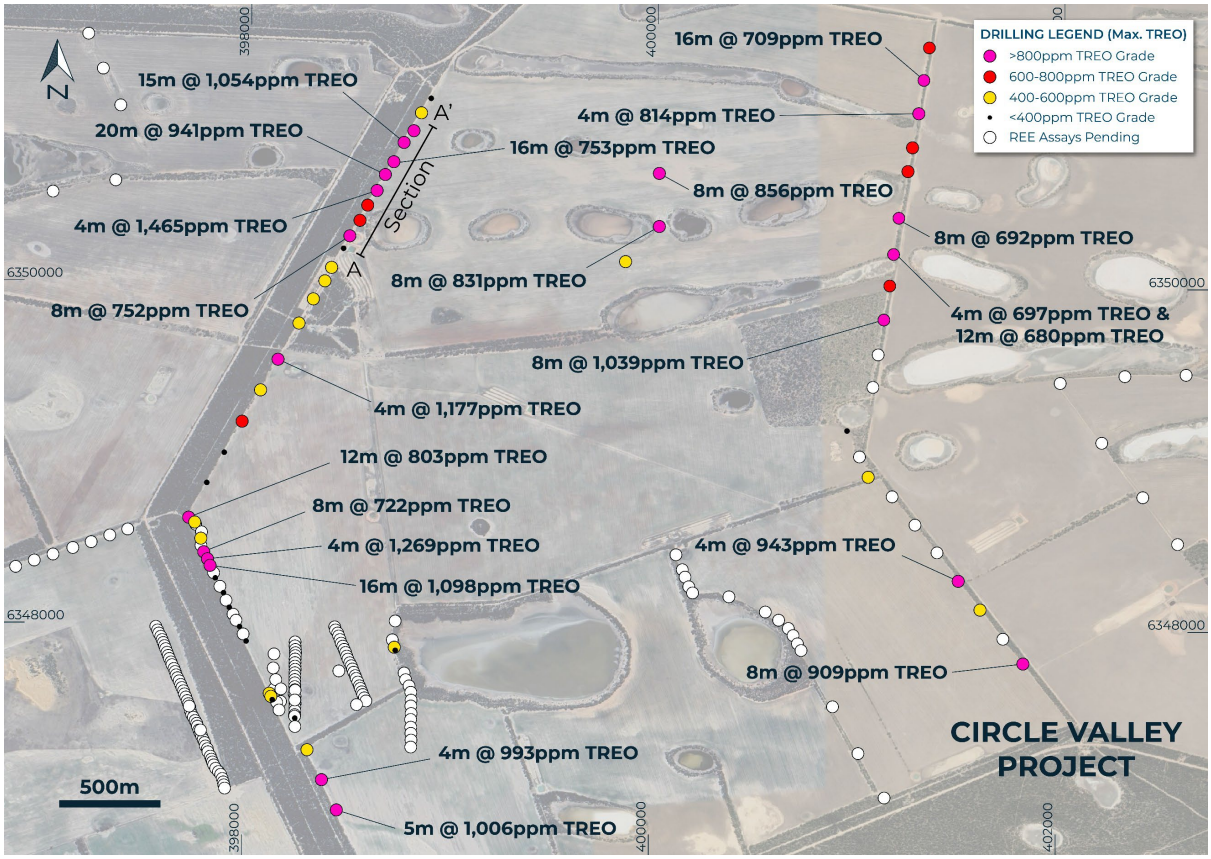


Figure 4: Central area showing consistent thick, high-grade rare earths at Circle Valley.

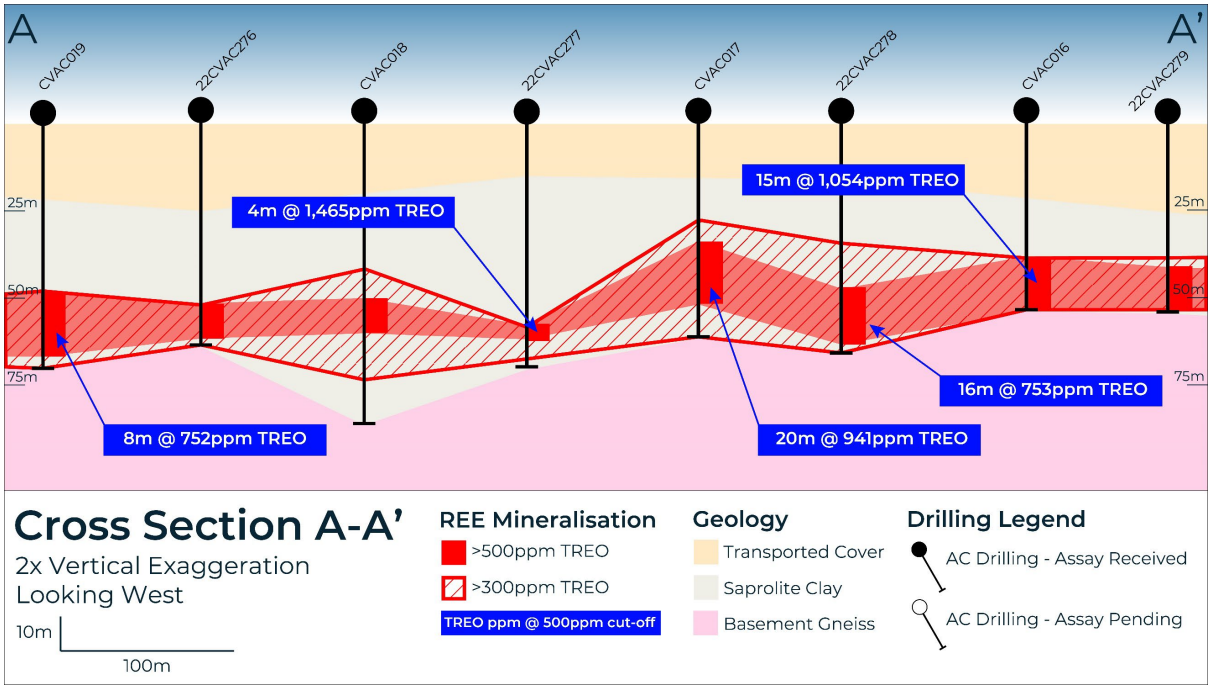


Figure 5: Section A-A' on Figure 4 – cross section through mineralisation.

ABOUT RARE EARTH ELEMENTS

The unique chemical and physical properties of rare earths have positioned them as a critical material across a number of rapidly evolving markets and industrial applications. In particular, Neodymium and Praseodymium oxides, which are critical elements in the manufacture of permanent magnets used for electric motors, turbines and mobile phones.

Key global megatrends are driving strong and diversified demand for Neodymium-Praseodymium oxides:



Low carbon energy transition – electric drive motors and turbines.



Military application – guidance and control systems.



Communications technology.



Sustainable resource security – increasing scarcity of and global competition for resources.



Supply chain security – against a backdrop of heightened geopolitical tension and push to diversify supply away from China.

To underscore the geopolitical importance of rare earths, the Pentagon has recently urged the US Government to fund Australian strategic mining of materials used to make electric vehicles and weapons. This is in an effort to reduce US reliance on China for lithium, rare earths and other minerals.

ABOUT CLAY HOSTED RARE EARTH DEPOSITS

Clay hosted rare earth deposits often enjoy significant project and cost advantages compared to hard rock deposits, with lower cost bulk mining and a simple process flow sheet. Clay deposits do not require the higher cost comminution and beneficiation processes that hard rock deposits require, resulting in lower capital intensity and lower operating cost to produce a refined product. The generally higher proportion of magnet rare earth elements (Neodymium-Praseodymium) in clay deposits also results in a high value product. Additionally, clay deposits may not produce the deleterious tailings waste.

Criteria	Clay Hosted REE	Hard Rock Hosted REE
Mineralisation 	▪ Elevated HREO/CREO.	▪ Can be either LREO or HREO dominant mineralisation.
Resource Definition 	▪ Rapid, shallow, drilling into clay. ▪ Lower cost.	▪ Slow, deeper, drilling into hard rock. ▪ Higher cost.
Mining 	▪ Shallow mining. ▪ Lower strip ratio. ▪ Higher productivity. ▪ No blasting required. ▪ Lower cost.	▪ Higher strip ratio. ▪ Lower productivity. ▪ Blasting required. ▪ Higher cost.
Processing 	▪ Simple process flow sheet. ▪ No comminution (crushing or milling). ▪ Lower capital and operating costs.	▪ Complex process flow sheet. ▪ Requires comminution and beneficiation. ▪ Higher capital and operating costs.
Environmental 	▪ Low levels of radionuclides. ▪ Non-radioactive waste. ▪ Progressive rehabilitation of mining footprint.	▪ Possible deleterious elements in waste.

FORTHCOMING ANNOUNCEMENTS

July – August 2022: Assays from the remaining 14,957m of drilling for high-grade rare earths at Circle Valley.

July – September 2022: Gold assays from Murchison Gold Project drilling.

July 2022: Quarterly Activity Report.

September 2022: Pre-feasibility Study for the Murchison Gold Project.

September 2022: Audited Annual Report.

October – December 2022: Rare earth assays from Cascade drilling.

This announcement has been authorised for release by the Company's Board of Directors.

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ABOUT MEEKA

Meeka Metals Limited is gold and rare earths company with a portfolio of high quality 100% owned projects across Western Australia.

Gold

Meeka's flagship Murchison Gold Project has a combined 343km^2 landholding in the prolific Murchison Gold Fields and hosts a large high-grade 1.1Moz JORC Resource. The Company is actively growing these Resources while also progressing toward production. The release of the Murchison Gold Project Scoping Study in December 2021 outlined a robust Project that produces over 420koz of gold.

In addition, Meeka owns the Circle Valley Project in the Albany-Fraser Mobile Belt (also host to the Tropicana gold mine – 3Moz past production). Gold mineralisation has been identified in four separate locations at Circle Valley and presents an exciting growth opportunity, which is being aggressively pursued.

Rare Earths

Meeka controls the Cascade Rare Earths Project (2,068km^2) in a region that is rapidly emerging as a highly prospective clay rare earths province. Importantly, the results to date contain high levels of permanent magnet metals being Neodymium-Praseodymium oxides. These metals are geopolitically critical, and Meeka intend to accelerate our understanding of Cascade by commencing initial metallurgical work. Furthermore, drilling will be ongoing.



Global Mineral Resource Summary

Project	Measured			Indicated			Inferred			Total		
	Tonnes ('000t)	Grade (g/t)	Ounces ('000oz)	Tonnes ('000t)	Grade (g/t)	Ounces ('000oz)	Tonnes ('000t)	Grade (g/t)	Ounces ('000oz)	Tonnes ('000t)	Grade (g/t)	Ounces ('000oz)
Andy Well	150	11.4	55	1,050	9.3	315	650	6.5	135	1,800	8.6	505
Turnberry				6,800	1.6	355	4,500	1.8	255	11,300	1.7	610
TOTAL	**150**	**11.4**	**55**	**7,850**	**2.7**	**670**	**5,150**	**2.4**	**390**	**13,100**	**2.6**	**1,115**

Notes:
1. Mineral Resources previously reported to the ASX on 18 May 2021 in announcement titled "Murchison Gold Mineral Resource Grows 44% to +1.1 Million Ounces". The Company is not aware of any new information or data that materially affects the information included in this announcement and that all material assumptions and technical parameters underpinning the estimates continue to apply and have not materially changed.
2. Mineral Resources are produced in accordance with the 2012 Edition of the Australian Code for Reporting of Mineral Resources and Ore Reserves (JORC 2012).
3. Andy Well Mineral Resource is reported using 0.1g/t cut-off grade.
4. Turnberry Open Pit Mineral Resource is reported within a A$2,400/oz pit shell and above 0.5g/t cut-off grade.
5. Turnberry Underground Mineral Resource is reported outside a A$2,400/oz pit shell and above 1.5g/t cut-off grade.

COMPETENT PERSON'S STATEMENT

The information that relates to Exploration Results as those terms are defined in the 2012 Edition of the "Australasian Code for Reporting of Exploration Results, Mineral Resources and Ore Reserve", is based on information reviewed by Mr Duncan Franey, a Competent Person who is a member of The Australasian Institute of Mining and Metallurgy and the Australian Institute of Geoscientists. Mr Franey is a full-time employee of the Company. Mr Franey has sufficient experience that is relevant to the style of mineralisation and type of deposit under consideration and to the activity being undertaken to qualify as a Competent Person as defined in the 2012 Edition of the 'Australasian Code for Reporting of Exploration Results, Mineral Resources and Ore Reserves'. Mr Franey consents to the inclusion in the report of the matters based on his information in the form and context in which it appears.

The information that relates to Mineral Resources was first reported by the Company in its announcement to the ASX on 18 May 2021. The Company is not aware of any new information or data that materially affects the information included in this announcement and that all material assumptions and technical parameters underpinning the estimates continue to apply and have not materially changed.

The information that relates to Scoping Study results is based on information compiled by Mr Tim Davidson, a Competent Person who is a Member of the Australian Institute of Mining and Metallurgy. Mr Davidson is a full-time employee of the company. Mr Davidson is eligible to participate in short and long-term incentive plans of and holds shares and performance rights in the Company as previously disclosed. Mr Davidson has sufficient experience in the study, development and operation of gold projects and consents to the inclusion in the report of the matters based on his information in the form and context in which it appears.

FORWARD LOOKING STATEMENTS

Certain statements in this report relate to the future, including forward looking statements relating to the Company's financial position, strategy and expected operating results. These forward-looking statements involve known and unknown risks, uncertainties, assumptions and other important factors that could cause the actual results, performance or achievements of the Company to be materially different from future results, performance or achievements expressed or implied by such statements. Actual events or results may differ materially from the events or results expressed or implied in any forward-looking statement and deviations are both normal and to be expected. Other than required by law, neither the Company, their officers nor any other person gives any representation, assurance or guarantee that the occurrence of the events expressed or implied in any forward-looking statements will actually occur. You are cautioned not to place undue reliance on those statements.